

term, of a given order in the Φ s, can only arise from terms in the bare superpotential of *higher* order in the Φ s.

$$C_1 < C_2$$

Here the R value $2(C_1 - C_2)$ of $\exp(2i\pi T)$ is *negative*, so positive powers of $\exp(2i\pi T)$ can compensate for the positive R values of Y and W_α , and $\mathcal{B}_\lambda$ may therefore contain terms of arbitrary order in Y and W_α . Using the chiral symmetry condition (29.3.34) and the R invariance condition

$$2 = \mathcal{N}_W + 2 \sum_r \mathcal{N}_r - 2a(C_2 - C_1), \quad (29.3.37)$$

we can, however, set limits on the structure of terms of a given order $\mathcal{N}_\Phi$ in the Φ s. Eqs. (29.3.34) and (29.3.37) have a trivial solution with $\mathcal{N}_r = 1$ for $r = \mathcal{N}_\Phi$, $\mathcal{N}_r = 0$ for other values of r , and $a = \mathcal{N}_W = 0$; this solution just represents the presence in $\mathcal{B}_\lambda$ of the original bare superpotential, with no radiative corrections. If there is no superpotential to begin with, then $\mathcal{N}_r = 0$ for all r , so Eq. (29.3.35) does not allow any terms in the Wilsonian Lagrangian with $\mathcal{N}_W = 0$, and so *no superpotential can be generated*. (In the supersymmetric version of quantum chromodynamics this conclusion is usually derived by noting that there is no possible term in the superpotential that would be consistent with all symmetries, but as we have now seen, the conclusion is much more general.) For renormalizable asymptotically free theories there is a useful limit on the structure of Φ -independent terms in $\mathcal{B}_\lambda$. The condition of renormalizability tells us that $\mathcal{N}_r = 0$ for $r > 3$, so by subtracting $2/3$ of Eq. (29.3.34) from Eq. (29.3.37) we find

$$2 \geq \frac{2}{3} \mathcal{N}_\Phi + \mathcal{N}_W + 2a \left(C_1 - \frac{1}{3} C_2 \right). \quad (29.3.38)$$

Asymptotic freedom requires that $3C_1 > C_2$, so for $\mathcal{N}_\Phi = 0$ (or $\mathcal{N}_\Phi > 0$) each term on the right-hand side is positive. It follows that we can have no Φ -independent terms of higher than second order in W , and these terms are also independent of T , so they again represent the one-loop contribution to the running coupling parameter τ_λ . But for $C_1 < C_2$ there is no general prohibition against terms of second or higher order in W and negative order in Φ .

29.4 Supersymmetry Breaking in Gauge Theories

We now turn to a question of great physical interest: in what gauge theories is supersymmetry spontaneously broken?